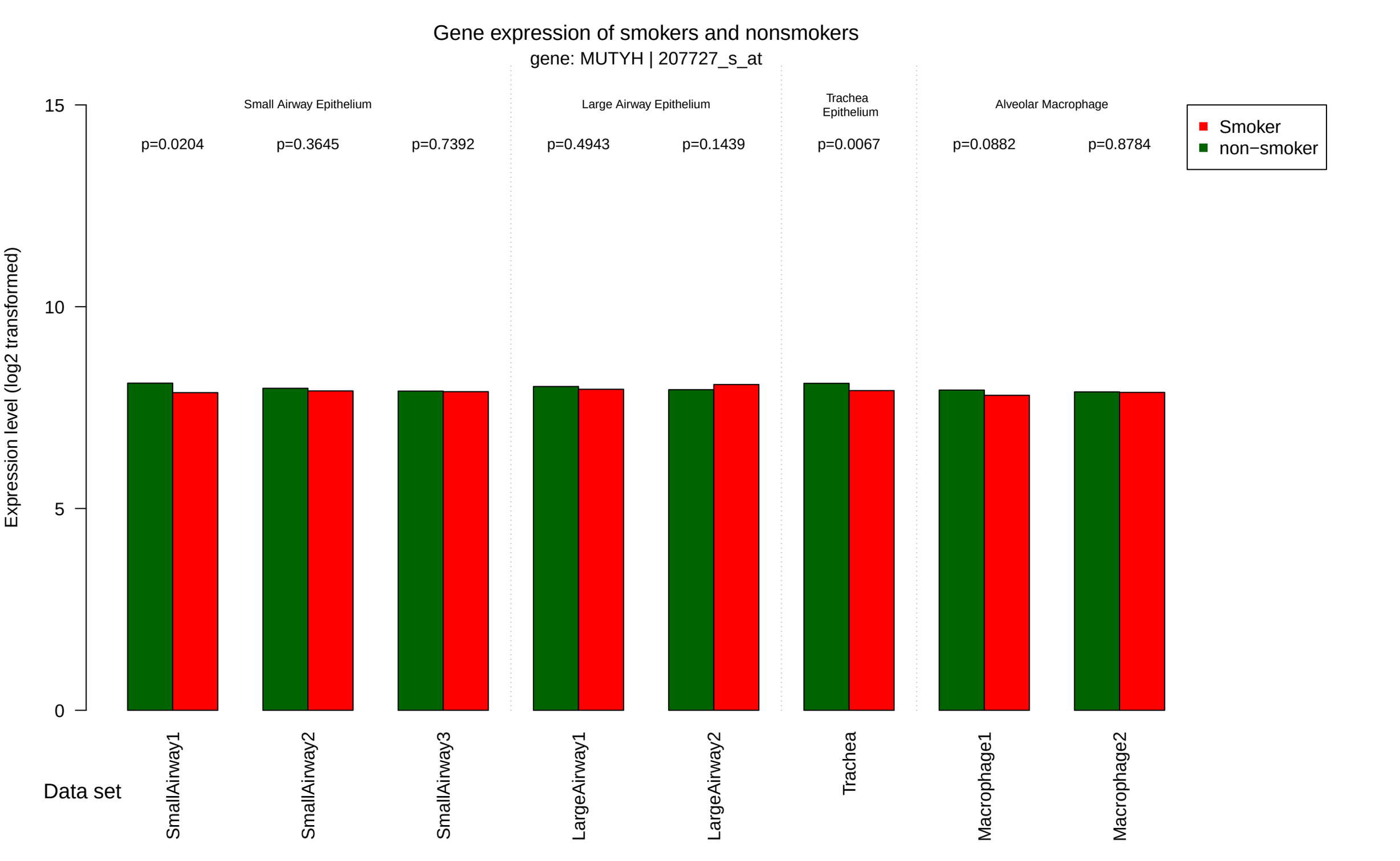
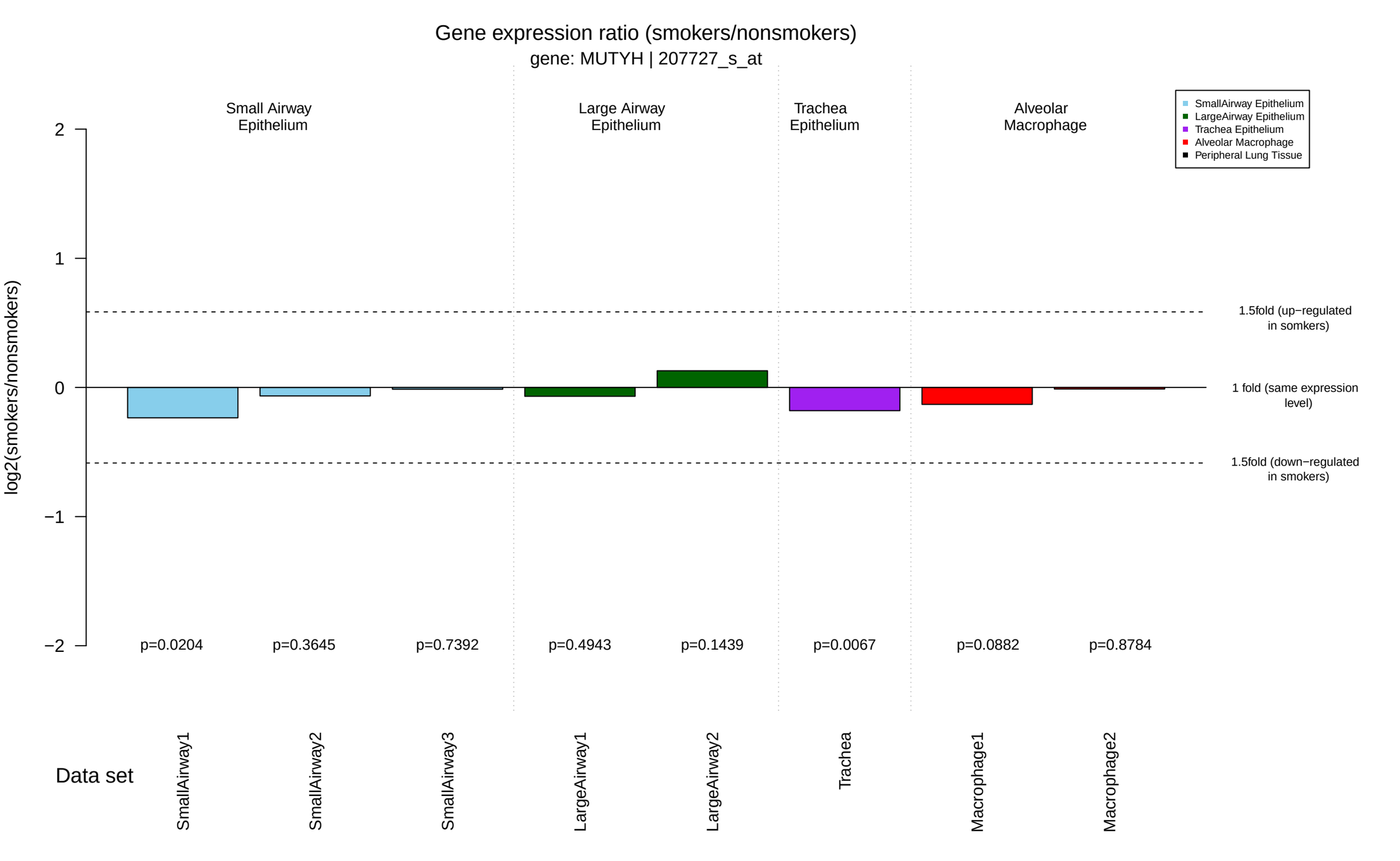
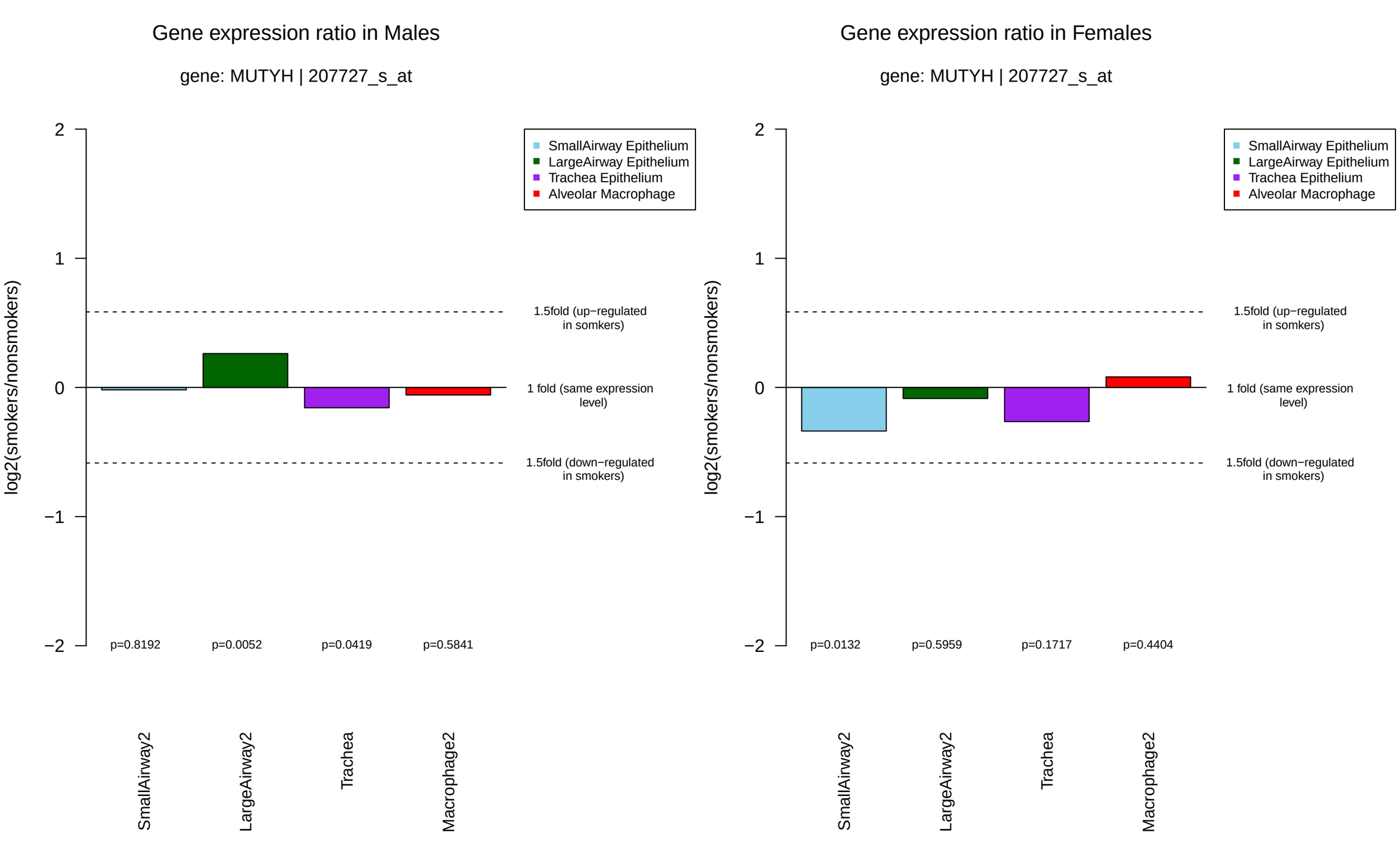
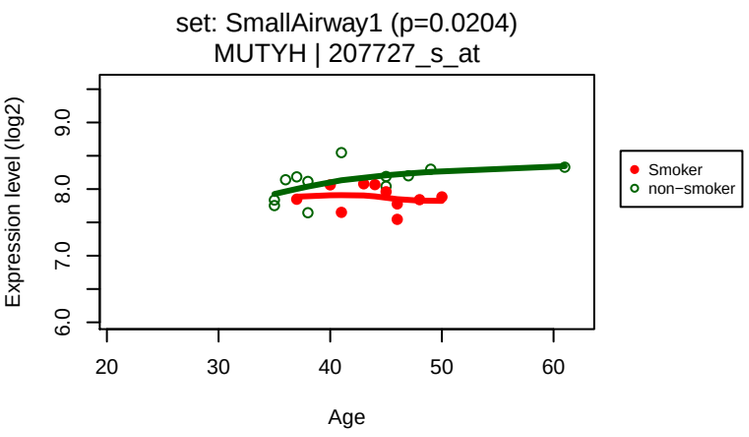
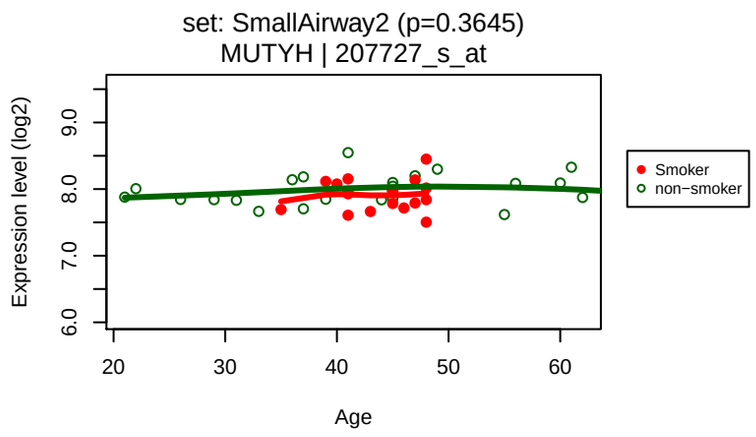
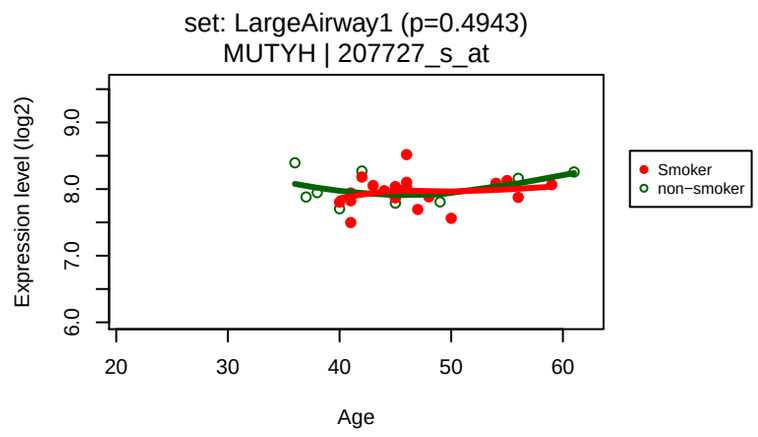

Gene expression of smokers and nonsmokers
gene: MUTYH | 207727_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0204
p=0.3645
p=0.7392
p=0.4943
p=0.1439
p=0.0067
p=0.0882
p=0.8784
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

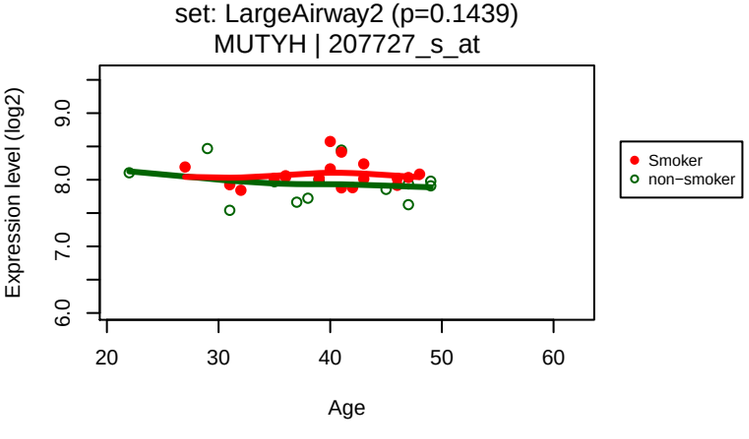

Gene expression ratio (smokers/nonsmokers)
gene: MUTYH | 207727_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0204
p=0.3645
p=0.7392
p=0.4943
p=0.1439
p=0.0067
p=0.0882
p=0.8784
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

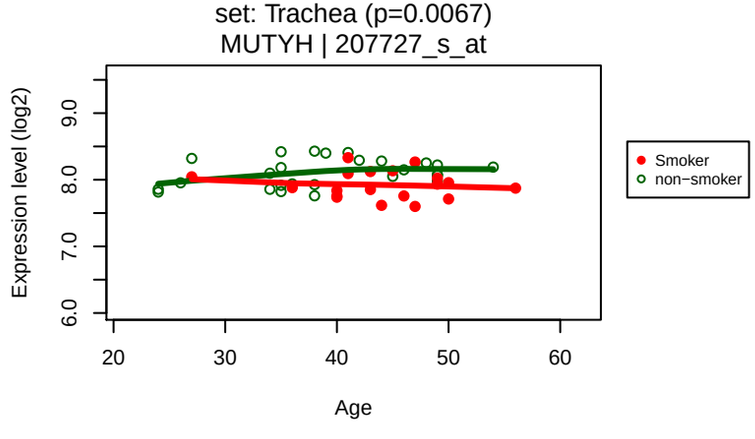

Gene expression ratio in Males
gene: MUTYH | 207727_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8192
p=0.0052
p=0.0419
p=0.5841
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUTYH | 207727_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0132
p=0.5959
p=0.1717
p=0.4404
SmallAirway2
LargeAirway2
Trachea
Macrophage2

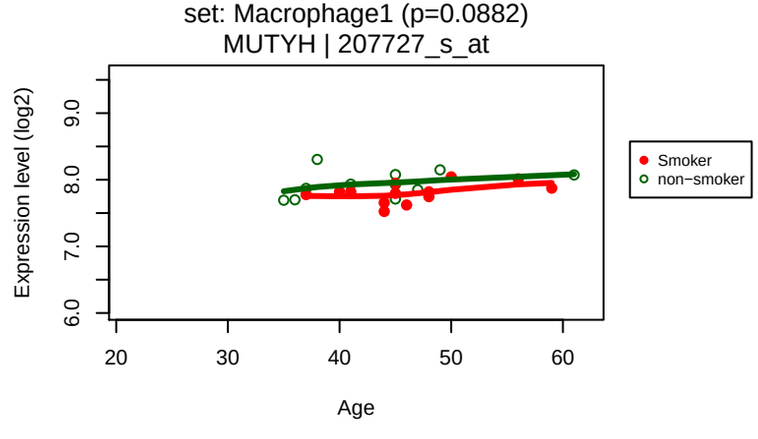

set: SmallAirway1 (p=0.0204)
MUTYH | 207727_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker

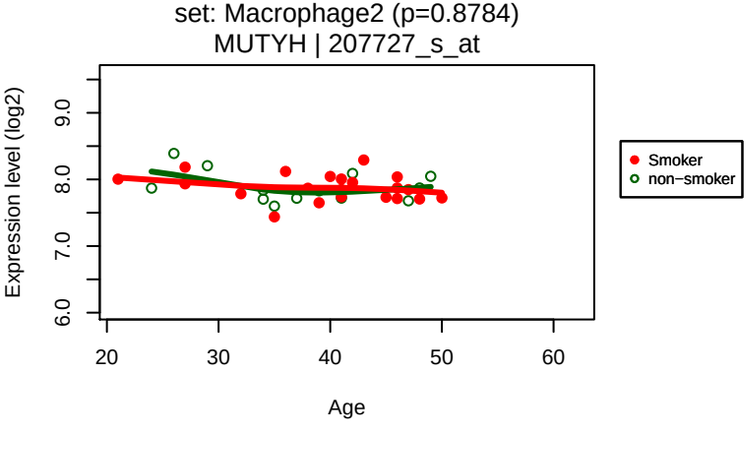

set: SmallAirway2 (p=0.3645)
MUTYH | 207727_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker

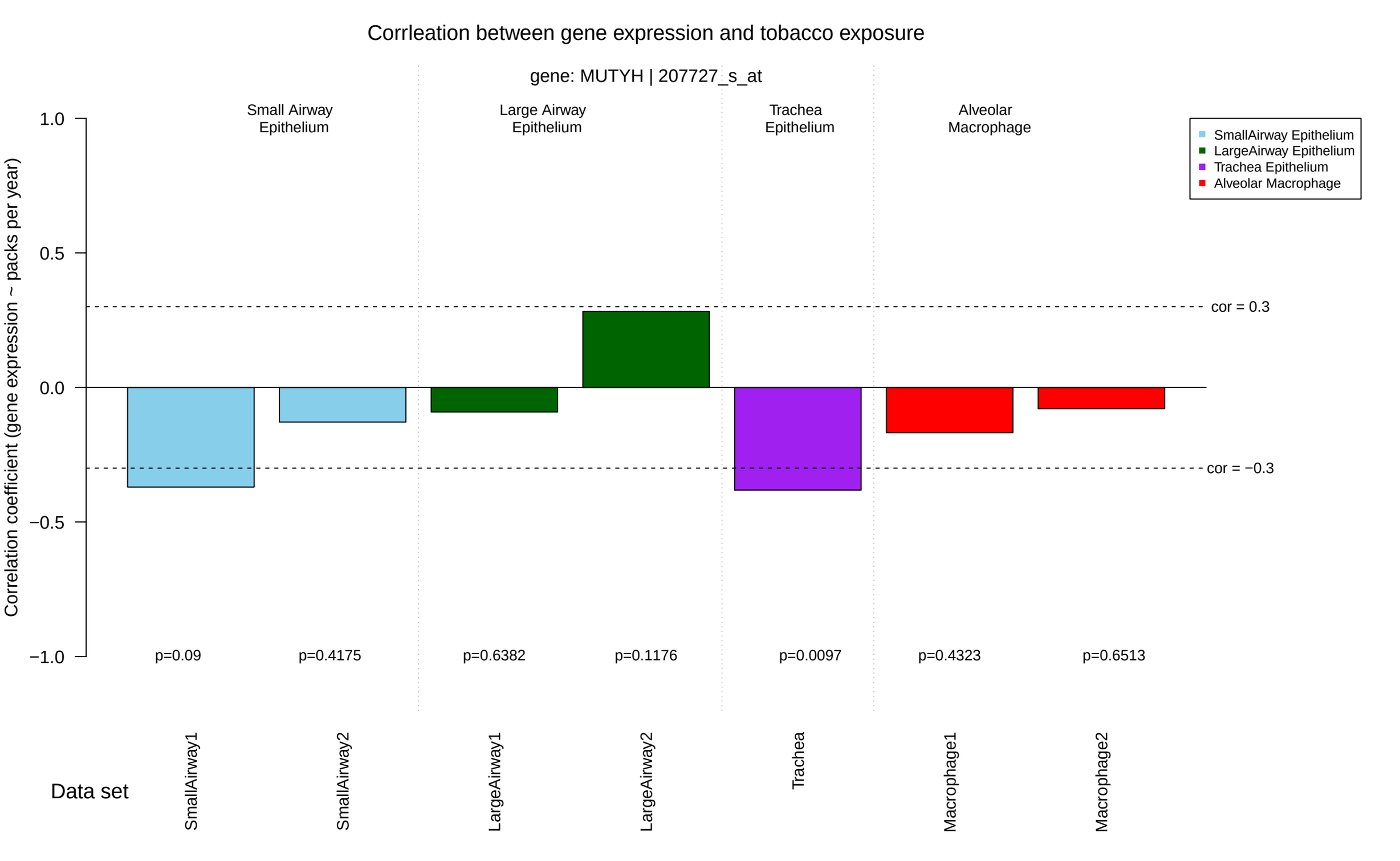

set: LargeAirway1 (p=0.4943)
MUTYH | 207727_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1439)
MUTYH | 207727_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0067)
MUTYH | 207727_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0882)
MUTYH | 207727_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8784)
MUTYH | 207727_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUTYH | 207727_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.09
p=0.4175
p=0.6382
p=0.1176
p=0.0097
p=0.4323
p=0.6513
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2